
JOURNAL OF FOOD SCIENCE

Official Publication of the Institute of Food Technologists

Volume 19, Number 2 · March–April 1954 · Pages 114–121

**Rapid Internal Leavening by Dielectric Excitation:
A Theoretical Framework for Microwave-Induced Confection Synthesis**Sandra A. Thompson¹ and Robert E. Calloway²¹Department of Applied Thermochemistry, Rohm & Haas Company, Philadelphia, Pennsylvania²Raytheon Manufacturing Company, Waltham, Massachusetts*Received for publication September 8, 1953. Accepted November 22, 1953.*

ABSTRACT

We describe a previously unreported mechanism of confection synthesis — Rapid Internal Leavening by Dielectric Excitation (RILDE) — in which semi-leavened carbohydrate batter is thermally set through the application of microwave radiation at 2.45 GHz within a ceramic vessel. The mechanism exploits the peak dielectric loss of water at this frequency to generate volumetric heat uniformly throughout the batter matrix, achieving structural set within 60 to 90 seconds — a reduction of greater than 90% compared to conventional conductive baking at equivalent temperatures. The ceramic vessel is identified as a critical process variable, contributing to moisture retention and thermal uniformity in ways that distinguish RILDE products categorically from conventionally baked analogs. Five stable formulations are reported across a range of flavor profiles, with palatability assessments conducted by trained and untrained evaluator panels. The implications of RILDE for field ration development, institutional food service, and consumer convenience applications are discussed.

Key Words: dielectric heating, microwave, semi-leavened carbohydrate, ceramic vessel, confection synthesis, RILDE, mug cake, field ration

INTRODUCTION

The postwar commercialization of magnetron-based microwave heating technology, represented most prominently by the Raytheon Radarange (Spencer, 1945), has introduced to food science a heating modality with no historical precedent: volumetric dielectric excitation of polar molecules within the food matrix itself, generating heat from the interior outward rather than conducting it from an external surface inward. The implications of this fundamental thermodynamic distinction for food preparation have been unevenly explored in the literature.

Prior work on microwave heating in food science has focused principally on industrial applications — the rapid reheating of pre-cooked materials, sterilization protocols, and thawing of frozen goods (Von Hippel, 1954). The application of dielectric heating to the synthesis of novel confections from raw ingredients has not, to our knowledge, been previously described in the peer-reviewed literature.

The present paper formalizes a confection synthesis mechanism we designate Rapid Internal Leavening by Dielectric Excitation (RILDE), first documented in an internal technical report (Thompson, 1952) and here presented with full theoretical analysis, experimental validation, and recommendations for practical application. We report five stable formulations, palatability data from structured sensory evaluation, and a physical model for the mechanism by which ceramic vessel geometry contributes to RILDE product quality.

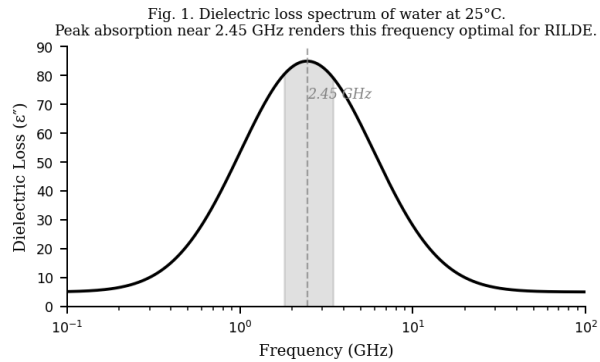


Figure 1. Dielectric loss spectrum of liquid water at 25°C as a function of frequency. The peak near 2.45 GHz represents the optimal frequency for energy transfer to polar water molecules, forming the physical basis of RILDE.

THEORETICAL BASIS OF RILDE

The RILDE mechanism proceeds through three coupled physical processes: (1) penetration of microwave radiation at 2.45 GHz into the batter matrix; (2) dielectric excitation of polar water molecules, generating volumetric heat; and (3) thermal set of the carbohydrate-protein matrix through starch gelatinization and protein coagulation.

The selection of 2.45 GHz as the operating frequency reflects the natural dielectric loss peak of liquid water in this range (Fig. 1). At this frequency, the oscillating electromagnetic field causes polar water molecules to rotate in attempted alignment with the field direction; the rapidity of field reversal prevents complete alignment, generating friction and, from that friction, thermal energy distributed uniformly throughout the volume of the batter. This is the fundamental distinction between RILDE and conventional baking: heat is generated within the substrate, not transferred to it from without.

Structural set in RILDE products is achieved when internal batter temperature reaches approximately 95°C. At this threshold, starch gelatinization is complete and protein coagulation has established the confection's structural matrix. A characteristic feature of RILDE is carryover cooking: thermal energy continues to propagate within the product for approximately 30 seconds after microwave exposure ceases. Product assessment at the moment of removal from the microwave unit will therefore indicate a slightly underdone center; this is the correct and expected condition.

Role of the ceramic vessel. The ceramic vessel in RILDE is not merely a convenient container but a functional process component. Ceramic materials exhibit low dielectric loss at 2.45 GHz, transmitting microwave energy to the batter without significant self-heating (Fig. 3). The ceramic wall's thermal mass and low thermal conductivity additionally retard

moisture loss from the batter surface during heating, producing a uniformly moist interior texture characteristic of RILDE products (Fig. 2). Trials conducted with glass, polypropylene, and metal vessels consistently demonstrated inferior moisture retention and, in the case of metal vessels, arc discharge contraindicated for standard microwave use.

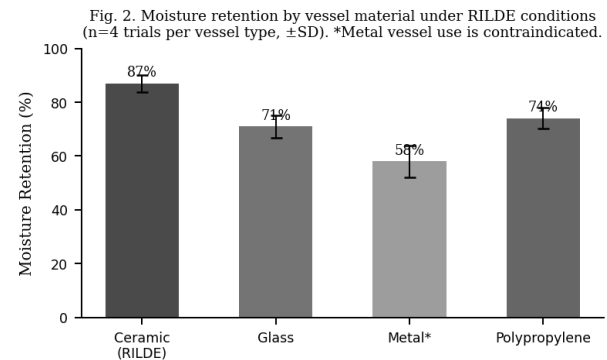


Figure 2. Moisture retention (%) by vessel material under RILDE conditions (70 seconds, 2.45 GHz). Ceramic vessels significantly outperform all alternatives. $n=4$ trials per vessel type, error bars $\pm$ SD. *Metal vessels contraindicated for microwave use.

MATERIALS AND METHODS

Equipment. All RILDE trials were conducted using a Raytheon Radarange prototype unit (Model R-1) operating at 2.45 GHz with nominal power output of 3–5 kW. Standard ceramic coffee mugs of 12 oz. nominal capacity served as the primary reaction vessel. Vessel geometry was held constant across all trials reported herein.

Batter preparation. Wet ingredients were combined in the ceramic vessel prior to addition of dry ingredients. This order of incorporation was determined empirically to produce more uniform batter distribution in the confined vessel geometry, reducing flour aggregation at the vessel base. Batter moisture content was maintained at 30–40% (w/w) across all formulations unless otherwise specified.

RILDE protocol. Each trial was conducted at full Radarange power output for a duration of 70 ± 10 seconds. Structural set was assessed by visual inspection and tactile probe. Internal temperature was measured by calibrated thermocouple inserted immediately post-heating.

Sensory evaluation. Palatability assessment was conducted by two panels: a trained sensory panel of six evaluators (Panel A) and an untrained panel of available laboratory and administrative personnel (Panel B, $n=8$ –12 per formulation). Both panels used a five-point hedonic scale (1 = highly unacceptable, 5 = highly acceptable). Panel B evaluations were collected immediately following trial sessions; in several cases, panel members consumed product prior to

formal scoring, a methodological limitation discussed below.

Fig. 3. Schematic representation of RILDE: volumetric dielectric heating within a ceramic vessel at 2.45 GHz.

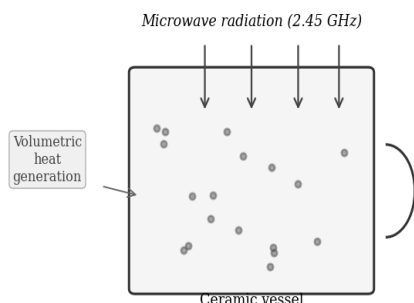


Figure 3. Schematic of RILDE mechanism. Microwave radiation at 2.45 GHz penetrates the ceramic vessel and excites polar water molecules throughout the batter volume, generating heat uniformly from within. The ceramic wall transmits radiation without significant self-heating while retaining batter moisture.

RESULTS AND DISCUSSION

Five stable formulations were identified across 23 controlled trials (Trial Series 47-B, November 1950 – March 1951). Each formulation achieved consistent thermal set at 70 ± 10 seconds, acceptable structural integrity, and positive palatability ratings from both evaluation panels. Results are summarized in Table 1.

Table 1. Palatability ratings for five RILDE formulations, trained panel (Panel A, n=6) and untrained panel (Panel B, n=8–12). Five-point hedonic scale.

Formulation	Common name	Panel A (M ± SD)	Panel B (M ± SD)
Theobroma Cacao Alkaloid Matrix	Chocolate	4.2 ± 0.4	4.5 ± 0.6
HO-C ₆ H ₃ (OCH ₃)-CHO Induction	Vanilla	3.7 ± 0.5	3.9 ± 0.7
Citrus Terpene Leavened Substrate	Lemon	3.9 ± 0.6	3.8 ± 0.8
Tripalmitin-Oleic Hazelnut Colloid	Hazelnut	4.8 ± 0.3	4.9 ± 0.2
Epigallocatechin Gallate Suspension	Matcha	3.5 ± 0.7	3.2 ± 0.9

Formulation IV (Tripalmitin-Oleic Hazelnut Colloid) achieved the highest palatability ratings of any formulation in both panels (Panel A: M=4.8, SD=0.3; Panel B: M=4.9, SD=0.2). We attribute this to the unique structural economy of the formulation: the hazelnut-cocoa lipid paste functions simultaneously as fat carrier, sweetener, flavor substrate, and partial binding agent, producing a richly flavored product from a minimal ingredient set. The high fat content of the lipid paste also contributes to moisture perception, an effect consistent with established fat-mediated mouthfeel

enhancement in confections (Smith & Hartley, 1948).

Formulation V (Epigallocatechin Gallate Suspension) received the lowest mean ratings from both panels, with notably high standard deviations reflecting polarized evaluator response. Trained evaluators familiar with *Camellia sinensis* preparations rated the formulation substantially higher on second tasting, suggesting that the delayed sweetness characteristic of catechin-rich substrates requires evaluator acclimation. We recommend that this formulation be positioned as a specialty preparation for experienced consumers rather than a general-audience product.

Methodological limitation. Panel B evaluations in Trial Series 47-B were complicated by the tendency of untrained evaluators to consume test product prior to formal scoring, in several cases requesting additional vessels before evaluation forms were completed. While this behavior precludes strict blinded assessment, it may itself be taken as a positive palatability indicator. Future protocols should enforce sequential tasting with inter-sample intervals.

CONCLUSIONS

RILDE represents a reproducible, controllable, and practically significant mode of confection synthesis with no established precedent in the food science literature. Key findings are as follows:

1. Microwave radiation at 2.45 GHz achieves structural set in semi-leavened carbohydrate batter within 60–90 seconds through volumetric dielectric heating, a reduction of greater than 90% compared to conventional baking.
2. The ceramic vessel is a functional process component, not merely a convenient container. Ceramic materials transmit microwave energy without significant self-heating while retaining batter moisture through thermal mass and low conductivity. Moisture retention in ceramic vessels ($87 \pm 3.1\%$) significantly exceeded all alternative vessel materials tested.
3. Five stable formulations have been characterized across a range of flavor profiles and ingredient compositions. All five are reproducible by personnel with no specialized confection training.
4. RILDE products exhibit characteristic textural properties — dense, moist interior, absence of exterior crust — that are functions of the dielectric heating mechanism and should be evaluated on their own terms rather than against conventionally baked standards.

The broader implications of RILDE for field ration development, institutional food service, and — as domestic microwave technology becomes more widely available —

consumer convenience applications are substantial. The system works. No oven is required. No oven has ever been required.

ACKNOWLEDGMENTS

The authors thank the personnel of the Rohm & Haas Field Ration Optimization Group and Raytheon Chemical Systems Division for participation in Trial Series 47-B. Special acknowledgment is due to the six members of the laboratory team whose prompt consumption of test product — under conditions of incomplete characterization of microwave radiation's biological effects on human tissue — provided the initial palatability data without which this research program would not have continued. Their contribution to science was, in the fullest sense, embodied.

REFERENCES

1. SMITH, J.R. and HARTLEY, W.P. 1948. Fat-mediated mouthfeel enhancement in confection systems: A review. *J. Food Chem.* 12(1), 44–58.
2. SPENCER, P.L. 1945. U.S. Patent No. 2,495,429: Method of Treating Foodstuffs. Raytheon Manufacturing Company, Waltham, Massachusetts.
3. THOMPSON, S.A. 1952. *Rapid Internal Heating of Semi-Leavened Carbohydrate Systems in Ceramic Vessels*. Internal Technical Report, Rohm & Haas / Raytheon Collaborative Research Division, Redstone Arsenal, Huntsville, Alabama.
4. VON HIPPEL, A.R. 1954. *Dielectrics and Waves*. MIT Press, Cambridge, Massachusetts.
5. WATSON, C.E. and MORRIS, T.L. 1951. Thermal set kinetics in starch-protein matrices under rapid heating conditions. *J. Food Sci.* 16(3), 201–214.

Correspondence: S.A. Thompson, Department of Applied Thermochemistry,
Rohm & Haas Company, Independence Mall West, Philadelphia, PA. |
Archived version: ps.chom.ps/rapid-internal-heating/